AUSTRONOMIC
Engineering the Cosmos

# Austronomic Weekly Journal

## Week 2: Light as Information

*Field Notes | Electromagnetic Foundations*
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**Abstract**

*Every piece of astronomical knowledge has arrived as light. This journal establishes light as a structured physical signal rather than passive illumination, and treats astronomy as the engineering discipline of preserving and decoding that signal across a sequential measurement chain. The wave-particle duality of electromagnetic radiation is examined from a systems perspective, distinguishing the wave regime that governs propagation and optical design from the particle regime that governs detection and noise. The electromagnetic spectrum is mapped as an information landscape: each waveband encodes a distinct class of astrophysical process, and atmospheric transmission windows define the boundaries of ground-based access. Four quantitative relations are derived and applied: the Doppler shift equation, which encodes source velocity in spectral line position; the photon energy relation, which sets the energy scale for every detector interaction; Wien's displacement law, which predicts the peak emission band of any thermal source; and the inverse-square law, which defines the fundamental sensitivity constraint of all astronomical instrumentation. These relations are unified in the Signal Chain Model, a seven-stage pedagogical framework that maps photon flow from source emission to detector output and serves as the structural backbone for the Austronomic series.*

**Keywords:** *astronomy education; electromagnetic spectrum; photon energy; Wien's displacement law; inverse-square law; Doppler shift; signal chain; observational astronomy; engineering pedagogy; Austronomic*

## 1. Introduction

Every piece of knowledge astronomers hold about the universe has arrived as light. Temperature, composition, velocity, and cosmic age are all encoded in photon properties that have travelled, sometimes for billions of years, to reach a detector. Light is not merely what makes astronomy possible; it is the only physical channel through which astronomical information can be received.

Treating light as illumination is inadequate. In an engineering framing, the sky is the transmitter, the atmosphere is a degrading medium, the telescope is the collection front end, and the detector is the sampler. Every signal property, wavelength, intensity, polarisation, and time variation, carries recoverable physical information. Astronomy is the discipline of preserving and decoding a signal that is always scarce and always degraded.

This journal maps the electromagnetic spectrum as an information landscape, derives the key quantitative relations governing photon energy, thermal emission, and geometric signal loss, and introduces the Signal Chain Model in Section 6 as a unified framework for all subsequent topics. By the end, the reader should understand light not just as a physical phenomenon, but as an engineering resource to be managed deliberately.

## 2. The Dual Nature of Light

Light is a transverse electromagnetic wave: oscillating electric and magnetic fields,

perpendicular to each other and to the direction of travel, propagating at constant velocity in vacuum. In its wave description, light is characterised by wavelength, frequency, amplitude, phase, and polarisation. In its particle description, it is a stream of discrete energy packets called photons, each carrying energy determined by its frequency. These are complementary descriptions of the same physical entity, and both are active in every astronomical observation [1].

The two descriptions govern different stages of the measurement chain. Wave optics determines how light propagates, diffracts through an aperture, and disperses through a spectrograph, setting angular resolution and spectral fidelity. Particle physics determines how light interacts with the detector: individual photon arrivals produce discrete electron liberation events, and the statistical uncertainty of counting those events is photon shot noise, the irreducible sensitivity floor of astronomical measurement. A larger aperture does not merely produce a brighter image; it produces a statistically more reliable one.

- Wave properties: wavelength, frequency, amplitude, phase, and polarisation state.
- Particle properties: energy per photon ($E = h\nu$), momentum, and discrete detection events.
- Wave behaviour dominates propagation and optical focusing.
- Particle behaviour dominates at the detector.

The transition from wave to particle description across the measurement chain is illustrated in Figure 1 [1, 2].

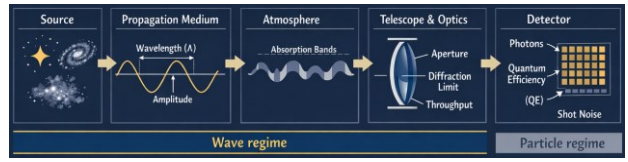


***Figure 1.*** *The astronomical signal chain from source to detector. Wave optics governs propagation and focusing; particle physics governs detection and noise. Both descriptions are simultaneously required for a complete system analysis.*

## 3. The Electromagnetic Spectrum as an Information Landscape

The electromagnetic spectrum spans roughly twenty orders of magnitude in frequency, from long-wavelength radio waves to the highest-energy gamma rays. Visible light, detectable by the human eye from approximately 380 to 700 nm, occupies a narrow slice near the middle of this range. Every band across the full spectrum is scientifically relevant, because different wavebands reveal different physical processes and different classes of astrophysical object [2, 3].

Wavelength is not a label for colour; it is a proxy for the energy scale of the emitting process. Thermal emission peaks at a wavelength set by the source temperature, a relation examined in Section 4. Non-thermal processes such as synchrotron radiation, inverse Compton scattering, and nuclear decay produce radiation at characteristic energies. Cold molecular clouds radiate in the microwave and far-infrared; stellar photospheres peak in the optical and near-infrared; plasma heated to millions of degrees emits X-rays and gamma rays. Each band is a window into a specific energetic regime.

Earth's atmosphere is not uniformly transparent and is opaque to most of the spectrum. Ground-based astronomy is confined to several distinct transmission windows. The optical window spans approximately 310 nm to about 1 μm, where ozone and molecular absorption define

the ultraviolet cutoff. Near-infrared windows in the J, H, and K photometric bands (approximately 1.1 to 2.5 μm) allow access to cooler thermal emitters. Mid-infrared windows in the N and Q bands (approximately 8 to 13 and 17 to 25 μm, respectively) are accessible only from high-altitude dry sites. The radio window extends from approximately 1 mm to tens of metres, bounded below by water vapour absorption and above by ionospheric reflection. All other wavebands require space-based instruments [2].

### Spectral Information Content

Beyond broadband intensity, a spectrum carries detailed diagnostic information. When atoms or molecules absorb or emit at specific quantised frequencies, they imprint characteristic lines on the spectrum. These lines reveal chemical composition, constrain temperature and pressure through broadening mechanisms, and expose radial velocity through wavelength shifts. A continuum encodes the overall energy budget and temperature; spectral lines encode the atomic and molecular physics of the emitting or absorbing material [3, 4].

A useful organising principle is to pose four questions of any astronomical source: what is it made of, how energetic is it, how is it moving, and how does it change with time? Composition is a spectral question. Energy scale is answered by continuum shape and the waveband of peak emission. Motion is encoded in wavelength shifts, quantified by Equation (1) below. Time variability reveals rotation, pulsation, eclipses, flares, and transients. This framework connects diverse observational techniques to a small number of physical objectives and will recur throughout this series.

The non-relativistic Doppler shift equation, relating the fractional wavelength displacement of a spectral line to the radial velocity of the source along the line of sight [3].

$$\frac{\Delta\lambda}{\lambda_0} = \frac{v_\mathrm{r}}{c} \qquad (1)$$

Equation (1) is the quantitative foundation of radial velocity astronomy. A spectral line shifted by one part in ten thousand corresponds to a radial velocity of approximately 30 km/s. The same equation, applied to entire galaxy spectra, underpins the measurement of cosmological redshift. It also illustrates the engineering consequence of spectral resolution: resolving a velocity of 1 km/s requires a spectrograph resolving power of roughly 300,000, placing strict demands on grating design and detector sampling. These values are examined quantitatively in Equation (1) [3]

$\boldsymbol{\Delta\lambda}$ = wavelength shift, in metres (m)

$\boldsymbol{\lambda_0}$ = rest wavelength of the spectral line, in metres (m)

$\boldsymbol{v_\mathrm{r}}$ = radial velocity of the source, in metres per second (m/s)

$\boldsymbol{c}$ = speed of light in vacuum, approximately 2.998 x 10^8 m/s

- Continuum shape encodes approximate temperature, luminosity, and distance via the inverse-square law.
- Absorption lines encode composition of cooler material along the line of sight, including stellar atmospheres.
- Emission lines encode temperature, density, and ionisation state of radiating plasma.
- Line shifts (Equation (1)) encode radial velocity and cosmological redshift; line widths encode turbulence and rotation.
- Polarisation encodes magnetic field orientation, scattering geometry, and aligned dust properties.

Table 1 summarises the five principal observables encoded in astronomical light, the technique most commonly used to measure each, and the physical information that each measurement recovers.

***Table 1.*** *Principal observables encoded in astronomical light and the physical interpretation each supports.*

| Observed Property of Light | Typical Measurement Technique | Physical Information Recovered |
|---|---|---|
| Wavelength pattern (spectrum) | Spectrograph or narrowband filter | Composition, ionisation state, temperature regime, line-of-sight absorption |
| Broadband flux and colour | Multi-band photometry | Approximate temperature, luminosity, reddening, variability amplitude |
| Time variation of flux | Repeated photometric or spectral measurements | Rotation, pulsation, eclipses, transits, flares, accretion events |
| Spectral line shift or width | High-resolution spectroscopy | Radial velocity, turbulence, expansion rate, rotational broadening |
| Polarisation state | Spectropolarimetry or imaging polarimetry | Magnetic field structure, scattering geometry, aligned dust |

## 4. Photon Energy and Thermal Emission

Two equations form the quantitative core of this section. Together they connect the wavelength of an observation to the energy scale of the physical process being probed, and predict the peak emission band of any thermally radiating body from its temperature. Each is a design input as much as a physics result: every detector choice and every instrument band selection consults these relations implicitly. The observing bands identified in Table 1 are each a direct consequence of these two equations [1, 3].

### The Photon Energy Relation

Because electromagnetic radiation is quantised into photons, the energy delivered per photon is determined entirely by its wavelength. This sets the energy scale for every interaction between light and matter, including detection. Equation (2) expresses this relation in its most compact form.

Photon energy as a function of frequency and wavelength, connecting the quantum particle description to the classical wave description [1].

$$E = hv = \frac{hc}{\lambda} \qquad (2)$$

Equation (2) explains why detector technology differs fundamentally across the spectrum. A single ultraviolet photon at 200 nm carries approximately 6.2 ev, enough to ionise hydrogen and drive photomultiplier or UV-sensitive Charge-Coupled Device (CCD) responses. A red photon at 700 nm carries about 1.8 ev, sufficient to liberate an electron in silicon. A microwave photon at 1 mm carries only about 0.001 ev, far below the threshold for photon counting, requiring bolometric or heterodyne detection instead. These values are order-of-magnitude estimates derived directly from Equation (2) [1]

$\boldsymbol{E}$ = energy of a single photon, in joules (J) or electronvolts (eV)

$\boldsymbol{h}$ = Planck constant, approximately 6.626 x 10^-34 J s

$\boldsymbol{v}$ = frequency of the radiation, in hertz (Hz)

$\boldsymbol{c}$ = speed of light in vacuum, approximately 2.998 x 10^8 m/s

$\boldsymbol{\lambda}$ = wavelength of the radiation, in metres (m)

### Wien's Displacement Law

A body at uniform temperature T emits a continuous spectrum described by the Planck function. The wavelength at which this spectrum peaks is given by Wien's displacement law, Equation (3). This relation is the primary tool for inferring stellar surface temperatures from colour measurements and for predicting the

observing band most sensitive to any thermal source.

Wien's displacement law: the peak emission wavelength of a blackbody is inversely proportional to its absolute temperature [1, 3].

$$\lambda_{\max} = \frac{b}{T} \tag{3}$$

Equation (3) provides immediate design guidance. A solar-type star at 5800 K peaks near 500 nm, in the optical. A cool brown dwarf at 1200 K peaks near 2.4 µm, largely invisible to a silicon CCD and requiring a near-infrared array. A protostellar dust envelope at 100 K peaks near 29 µm, demanding a mid-infrared space telescope. These are approximate values computed via Equation (3) [1, 3].

$\boldsymbol{\lambda_{\max}}$ = wavelength at peak emission, in metres (m)

$\boldsymbol{b}$ = Wien's displacement constant, approximately 2.898 x 10^-3 mK

$\boldsymbol{T}$ = absolute temperature of the blackbody, in kelvin (K)

Table 2 applies Equation (3) to six representative astrophysical thermal sources, showing how peak wavelength and required observing band vary across twelve orders of magnitude in temperature.

***Table 2.*** *Approximate peak emission wavelengths for representative astrophysical thermal sources, computed via Equation (3). All values are order-of-magnitude estimates. The observing band column reflects the atmospheric window constraints described in Section 3.*

| Source Type | Approx. Temperature (K) | Peak Wavelength (approx.) | Primary Observing Band |
|---|---|---|---|
| Hot O-type star | 40,000 | 72 nm | Ultraviolet (space required) |
| Sun-like G-type star | 5,800 | 500 nm | Optical (green) |
| Cool M-type red dwarf | 3,500 | 828 nm | Near-infrared |
| Brown dwarf (T class) | 1,200 | 2.4 um | Near-infrared (K band) |
| Protostellar dust envelope | 100 | 29 um | Mid-infrared (space required) |
| Cold molecular cloud | 15 | 193 um | Submillimetre (high-altitude or space) |

## 5. The Inverse-Square Law

The observing band constraints visible in Table 2 tell us where to point an instrument. The question of whether a source is bright enough to detect from that distance is answered by the inverse-square law. As electromagnetic radiation propagates outward from a compact source, its total power is conserved but spread over an ever-expanding spherical surface. The power intercepted by any fixed collecting area falls as the square of the distance. This is an inescapable consequence of three-dimensional geometry that applies to every isotropic source in the absence of intervening absorption.

The practical consequence is severe: doubling the distance to a source reduces received intensity by a factor of four. Detecting a source ten times more distant at the same signal quality requires, in principle, one hundred times more collecting area or one hundred times longer integration. This is why aperture is the dominant sensitivity lever in astronomical instrumentation, and why the history of telescope development is largely a history of increasing primary diameter [3].

This geometric relationship is expressed formally as Equation (4), which gives the received intensity at any distance from an isotropic point source of known luminosity [3].

The inverse-square law for received intensity from an isotropic point source, relating

luminosity and distance to the flux arriving at the observer [3].

$$I = \frac{L}{4\pi d^2} \tag{4}$$

Equation (4) is the starting point for every sensitivity estimate in telescope and instrument design. Given a source luminosity and distance, it yields the intensity at the top of the atmosphere. Multiplied by collecting area, atmospheric transmission (typically 0.70 to 0.90 in the optical at a good site), and system throughput (typically 0.30 to 0.70 for the full optical train), it yields the approximate photon count rate at the detector. Every observational feasibility calculation begins with this expression, as formalised in Equation (4) [3]

$\boldsymbol{I}$ = received intensity (irradiance), in W/m^2

$\boldsymbol{L}$ = source luminosity (total emitted power), in watts (W)

$\boldsymbol{d}$ = distance from source to observer, in metres (m)

$\boldsymbol{4\pi d^2}$ = surface area of a sphere of radius d, in m^2

## 6. The Signal Chain Model

The Signal Chain Model is the central pedagogical framework of this journal series. It treats the flow of photons from an astronomical source to a recorded measurement as a sequential chain of seven stages, each of which modifies the signal in a physically definable way. An explicit accounting of this chain is what separates disciplined observational planning from equipment-driven intuition. The four equations introduced in Sections 3 through 5 each correspond to a specific stage in this chain.

Stage 1 is source emission: the photon flux leaving the source, set by source physics and captured by Equation (4). Stage 2 is interstellar propagation, where dust and gas along the line of sight absorb or scatter photons in a wavelength-dependent manner. Stage 3 is atmospheric transmission, which confines ground-based access to the windows described in Section 3. Stage 4 is telescope collection, where aperture area determines how many photons enter the system. Stage 5 is optical throughput, the combined transmission of all mirrors, lenses, filters, and dispersive elements. Stage 6 is detector conversion, where photon energy is converted to a measurable electronic signal with an efficiency called quantum efficiency. Stage 7 is noise, where shot noise, sky background, read noise, and thermal dark current combine to set the minimum detectable signal. At every stage, information can be irreversibly lost.

The seven stages of the Signal Chain Model and their relationships are shown schematically in Figure 2 [2, 3].

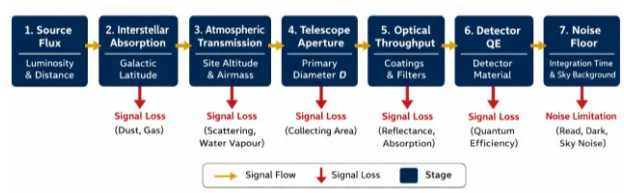


***Figure 2.*** *The Signal Chain Model: seven sequential stages from source emission to noise floor. Each stage modifies the photon budget in a physically definable way. Stages 2 through 7 each represent a potential site of signal loss or information degradation.*

Table 3 maps all seven stages of the Signal Chain Model to their primary physical process, a typical efficiency or loss value, the key controlling parameter at that stage, and the week in this series where each stage is examined in depth.

***Table 3.*** *The Signal Chain Model: seven stages mapping photon flow from source to measurement. Efficiency ranges are approximate and typical of optical ground-based systems. Each stage corresponds to one or more subsequent weeks in the Austronomic series.*

| Signal Chain Stage | Primary Physical Process | Typical Value or Loss (approx.) | Key Controlling Parameter | Week Examined in Depth |
|---|---|---|---|---|
| 1. Source flux | Inverse-square law plus source spectrum | Input (not a loss stage) | Source distance, luminosity, wavelength | W2 (this week) |
| 2. Interstellar absorption | Dust scattering and absorption | Variable; up to several magnitudes | Galactic latitude, source distance | Series Part 2 |
| 3. Atmospheric transmission | Molecular absorption, Rayleigh scattering | Approx. 0.70 to 0.90 in optical | Site altitude, airmass, precipitable water | W6 |
| 4. Telescope collecting area | Aperture geometry | Set by primary diameter | Primary aperture diameter D | W3 |
| 5. Optical throughput | Reflection, refraction, filter bandpass | Approx. 0.30 to 0.70 overall | Number of elements, coatings, filter width | W4, W5, W11 |
| 6. Detector quantum efficiency | Photoelectric conversion | Approx. 0.50 to 0.95 (silicon, optical) | Detector material, design, temperature | W8 |
| 7. Noise floor | Shot noise, sky background, read noise | Determines achievable SNR | Integration time, detector noise, sky brightness | W9 |

## 7. Why Astronomy is Fundamentally Signal-Limited

The Signal Chain Model in Figure 2 and Table 3 makes the source of this constraint visible: at every stage from source to detector, signal is lost or degraded, and no stage can recover information that was discarded upstream. Astronomical measurement therefore operates permanently under a signal scarcity constraint imposed by three simultaneous physical laws: the inverse-square law of Equation (4) reduces source intensity with distance; quantum detection statistics impose shot noise that is irreducible; and atmospheric opacity restricts spectral access to the narrow windows described in Section 3. No improvement in technology removes any of these constraints. They are structural features of the discipline, and every instrument design is a negotiated response to their combined effect.

Understanding how these constraints interact is more useful than understanding them in isolation. A decision to observe at shorter wavelengths improves angular resolution but increases photon energy, which for a given source flux reduces the photon count rate. Widening the spectral bandpass collects more photons but dilutes spectral information content. Increasing aperture improves both sensitivity and resolution, but scales cost and mechanical complexity steeply. Each design choice trades one resource against another, and the Signal Chain Model introduced in Section 6 is the accounting tool that makes those trades explicit and quantitative.

- Shot noise scales as the square root of detected photon count: signal-to-noise ratio grows only as the square root of integration time or collecting area.
- Sky background imposes a brightness floor on ground-based imaging that limits source detectability regardless of exposure time.
- Atmospheric opacity restricts spectral access, requiring space platforms for ultraviolet, far-infrared, and most X-ray science.
- Diffraction sets a wavelength-dependent angular resolution limit, linking spatial information content to aperture diameter.

This is precisely why the Austronomic series begins with light rather than with equipment. Any instrument specification only makes sense when it can be traced to a physical requirement on the number, energy, and wavelength distribution of photons that must be collected to achieve a defined measurement objective. Light

is not the background context for the equipment discussion; it is the engineering specification from which all equipment requirements follow [2, 3].

## 8. Summary and Bridge to Week 3

This journal has established light as a structured physical signal and astronomy as the engineering discipline of preserving and decoding it. The key results are as follows. Equation (1) encodes source radial velocity in spectral line shifts, telling us that a spectrograph resolving power of 300,000 is required to measure velocities at the 1 km/s level. Equation (2) links photon energy to wavelength, explaining why ultraviolet photons ionise detectors while microwave photons require bolometric receivers. Equation (3) predicts the peak emission band of any thermal source, showing that a 100 K dust cloud demands a mid-infrared space telescope while a 40,000 K star is best observed from space in the ultraviolet. Equation (4) quantifies geometric signal loss with distance, establishing aperture as the primary sensitivity lever and making every telescope size decision a direct consequence of the sources being targeted. The Signal Chain Model unifies these results into a seven-stage accounting framework that will be elaborated stage by stage throughout this series.

Week 3 takes the next step in the chain. Week 2 has established what information exists in light and how signal strength falls with distance. Week 3 addresses how much of that information an aperture can collect and spatially resolve. The primary mirror or lens diameter determines not only the photon count rate via collecting area but also the angular resolution limit via diffraction. The diffraction equation will be derived formally, and its implications for every telescope architecture will be made explicit in terms of the Signal Chain framework introduced here.

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## Author Biography

**Farzad Farajizadeh (Farzad Faradjizadeh)** is an electrical engineer with a doctoral background and a multidisciplinary technical foundation spanning power systems, power electronics, electromagnetic compatibility, and related engineering domains. Through Austronomic, he explores astronomy from a systems-oriented perspective, with a particular emphasis on connecting physical principles, instrumentation, and engineering reasoning in a form accessible to motivated learners.

His broader objective with Austronomic is to develop a structured pathway into astronomy that moves beyond equipment-driven decision making and instead begins with conceptual clarity, technical rigor, and long-term system thinking. This work reflects his interest in building an educational framework in which astronomy can be approached not only as an observational pursuit, but also as a disciplined engineering and scientific practice.

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